

Machine Judges Reduce Sentencing Bias? A Computational Social Science Evaluation

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Introduction

Questions

ML performs better than humans in classification tasks in CJ systems. Theoretically, ML has an ability to produce less-biased results (Kleinberg, Lakkaraju, Leskovec, Ludwig, & Mullainathan, 2017; Goel, Rao, & Shroff, 2016; Dressel & Fraid, 2018; Lin, Jung, Goel, & Skeem, 2020; Hu, Huang, Li, & Lu, 2025). What about in regression tasks? Evaluating models' fairness should be essential for court's decision on choosing ML to provide suggestions.

Machines and Biases

Research Gaps

- ▶ all focus on classification tasks
- ▶ no suitable tools to identify sentencing tasks performance (all based on causal inference)
- ▶ machine really outperform humans? lack mechanism analysis

Methodology

Data

all CJO theft sentencing documents data ranging from 2018 to 2021.

- ▶ 238,419 cases, 159,699 for model, 78,720 for validation
- ▶ only first-trial, one criminals
- ▶ regulation expressions (Li, Qin, & Yu, 2022; Xiao, Hu, Liu, Tu, & Sun, 2021), LawFormer (Xiao et al., 2021), Qwen 3, legal students assistants.

Methodology

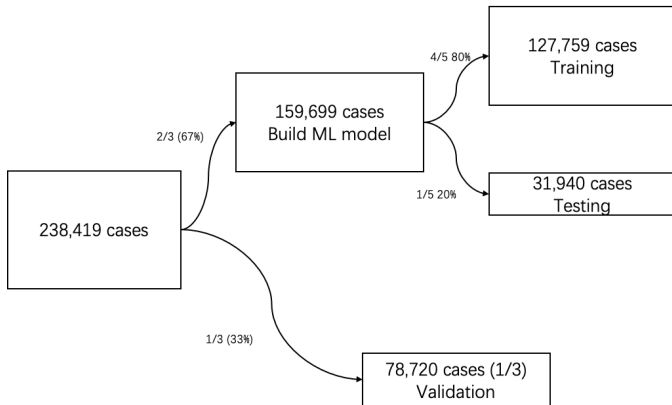


Figure 1: Illustration of Data Splitting

Methodology

Machine Learning Models

XGBoost is used and it produce similar results to LightGBM model:

$$f(x) = \sum_{m=1}^M \gamma_m h_m(x)$$

loss function:

$$\ell(y, f(x)) = \frac{1}{2}(y - f(x))^2$$

L2 regularization:

$$\Omega(f) = \lambda \sum_{m=1}^M \|h_m\|^2$$

CAIL2018 score: 80.1; MAE: 3.406. A very excellent model compared with Xiao et al. (2018) and Liu and Li (2019).

Methodology

Method: Macro Level

cosine similarity:

$$CS_{A,B} = \frac{X_A \cdot X_B}{\|X_A\| \|X_B\|} = \frac{\sum_{i=1}^n x_{A,i} x_{B,i}}{\sqrt{\sum_{i=1}^n x_{A,i}^2} \sqrt{\sum_{i=1}^n x_{B,i}^2}}$$

sentencing similarity:

$$SS_{A,B} = |I_A - I_B|$$

relative ratio:

$$RR = \frac{\frac{b_t}{b_t + u_t}}{\frac{b_c}{b_c + u_c}}$$

Methodology

* NOTE: This procedure should be conducted for n times

All cases

Case number	Legal factor 1	Legal factor 2	...	Legal factor n	Sentencing
1	0/1	0/1	...	0/1	S1
2	0/1	0/1	...	0/1	S2
...	...	...	...	...	...
n	0/1	0/1	0/1	0/1	Sn

Extract the baseline case
take No. 1 case as an example

Case number	Legal factor 1	Legal factor 2	...	Legal factor n	Sentencing
1	0/1	0/1	...	0/1	S1
2	0/1	0/1	...	0/1	S2
...	...	...	...	...	...
n	0/1	0/1	0/1	0/1	Sn

Cosine Similarity between all cases and the baseline case

Euclidean Distance between all cases and the baseline case

Change baseline and repeat this procedure

baseline: 1
Dependent variable:
number/Victim[1]
number/Reason[1]
Constant
Observations
R ²
Note: *p<0.1, **p<0.05, ***p<0.01

< 0: no bias
≥ 0: bias exists

Build regression on Cosine Similarity and Euclidean Distance

Case number	Cosine similarity	Euclidean Distance
2	?	?
...	...	...
n	?	?

Independent variable

Dependent variable

After calculating, delete the baseline

Figure 2: Method to Identify Biases based on Similarity

Methodology

Method: Micro Level

Compare each judge with machine.

Results: Macro

Table 2 Machine v.s. Judges: Strategy One

	Biased	Unbiased	Total	Inc risk ¹ /RR/OR (95% CI)
Machine Judges	8644	70076	78720	10.98 (10.76 to 11.20)
Human Judges	13217	65503	78720	16.79 (16.53 to 17.05)
RR				0.65 (0.64 to 0.67)
OR				0.61 (0.59 to 0.63)

Uncorrected chi2 test that OR = 1: $\text{chi2}(1) = 1110.849$, $\text{Pr} > \text{chi2} = < 0.001$

¹Outcomes per 100 cases units.

Results: Macro

Table 3 Machine Modified Human Judges v.s. Judges: Strategy Two

	Biased	Unbiased	Total	Inc risk ¹ /RR/OR (95% CI)
Modified Judges	5992	72728	78720	7.61 (7.43 to 7.80)
Human Judges	13217	65503	78720	16.79 (16.53 to 17.05)
RR				0.45 (0.44 to 0.47)
OR				0.41 (0.40 to 0.42)

Uncorrected chi2 test that OR = 1: $\text{chi2}(1) = 3205.928$, $\text{Pr} > \text{chi2} = < 0.001$

¹Outcomes per 100 cases units.

Results: Macro

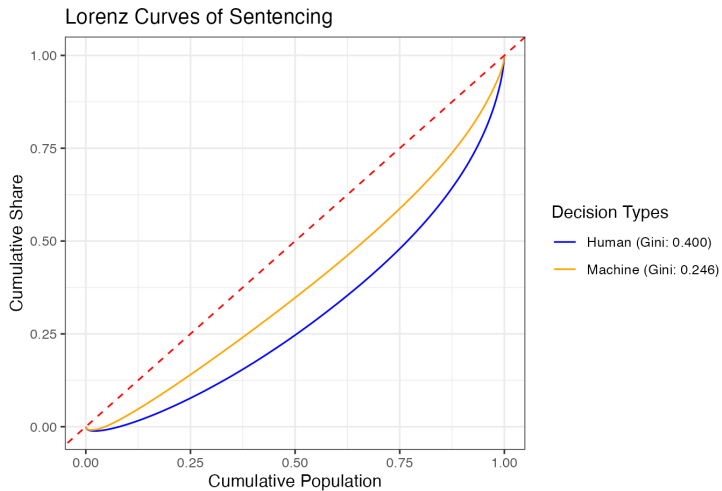


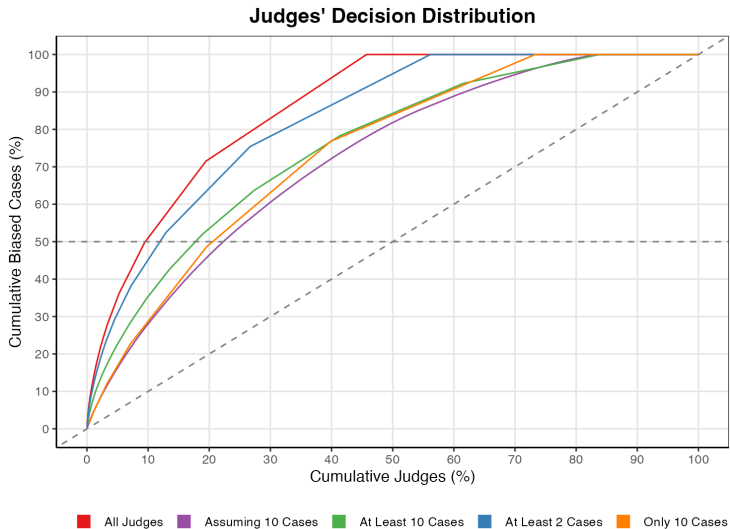
Figure 3: Lorenz Curve

Results: Micro

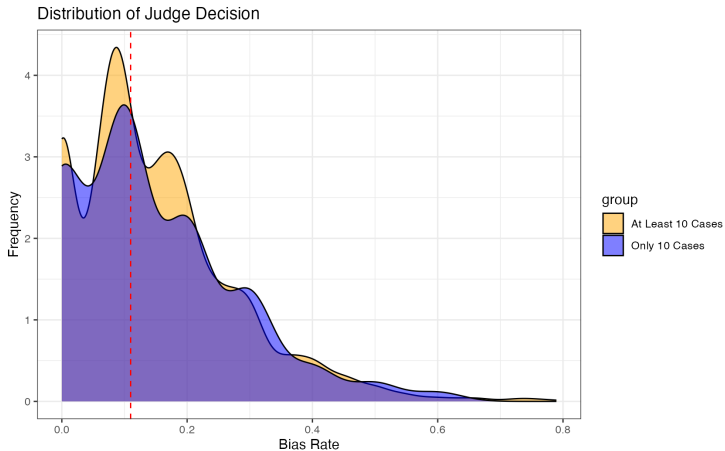
Table 2. Comparison Results: Micro View

	Worse Judges	Better Judges	Total Judges	Proportion (Worse)
All Judges	5059	7336	12395	0.408
Two Cases	4469	4572	9041	0.494
Ten Cases	1078	935	2013	0.536
Only Ten	112	169	281	0.399

Results: Micro



Results: Micro



Conclusions

Findings

- ▶ if all judges are replaced by machine learning models, the probability of being sentenced an unfair result is 35% lower
- ▶ if cooperating with judges, 55% biased cases can be sentenced in a more fair way
- ▶ extreme terrible judges contribute to the previous results. Only half judges perform worse than machine.

Thanks for your attention!

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